

User Manual

Application Introduction

Welcome to our virtual reality (VR) educational game --- Critter Escape: VR Adventure!

Our application is designed for students who are interested in VR and use VR technology to learn through mini-games.

By playing our game, you will not only improve your cognitive skills and science, technology, engineering, and mathematics (STEM) knowledge but also enhance your teamwork spirit.

The flow of the game is to cooperate with a partner to solve five different problems and escape from the dungeon. The game contains lots of STEM knowledge, including biology, arithmetic, and geometry.

Through playing the game, you can gain the joy of substituting a game role, and deepen the friendship with your friend.

This game provides a comfortable experience without dizziness and is friendly to any level of VR user, whether you are a beginner or an experienced professional.

Now let's begin your adventure in VR!

Prerequisites

To ensure a seamless and immersive gaming experience, please ensure the following requirements are met:

Hardware Requirements:

- **VR Headsets and Controllers:** A pair of Meta Quest 3 headsets and controllers are required to experience the full immersive capabilities of our game and ensure compatibility.
- **Personal Computer (PC):** Two Windows 10/11 PCs equipped with an RTX 2050 GPU or above are required to ensure optimal performance and graphics capabilities for a good game experience

Software Requirements:

- **VR Platform:** Oculus or other OpenXR-compatible platforms to connect PCs with headsets to play our game. There is a tutorial to install the Oculus PC app for Meta Quest Link: <https://www.meta.com/help/quest/articles/headsets-and-accessories/oculus-rift-s/install-app-for-link/>

Others:

- **Network:** As a multiplayer game, a robust Wi-Fi network environment is required.
- **USB Connectivity:** Two USB 3.0 (or above) cables are required for connection between your PCs and headsets.

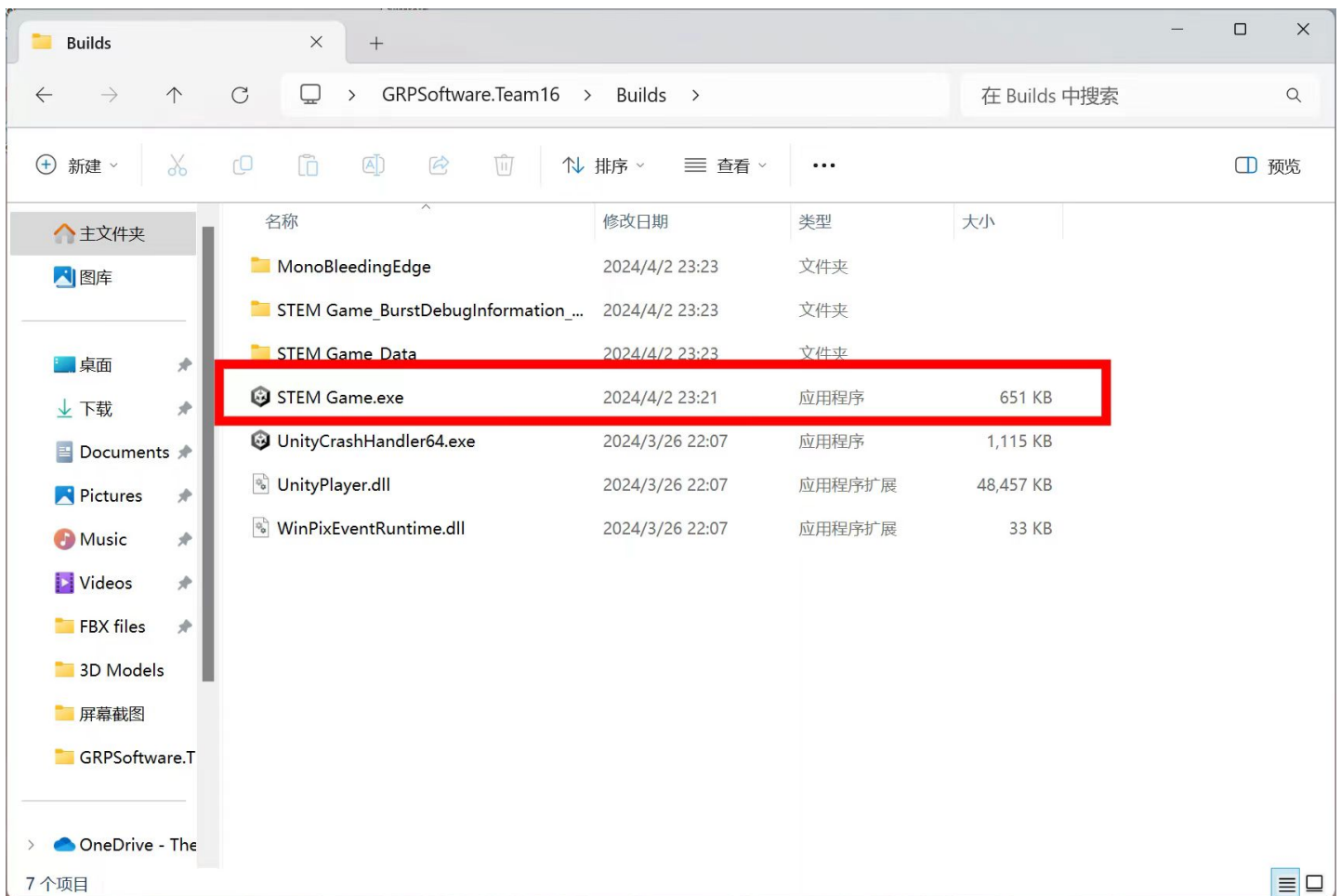
About this Application

- Current Version: 1.7.0
- Permission: No need

Installation Instructions

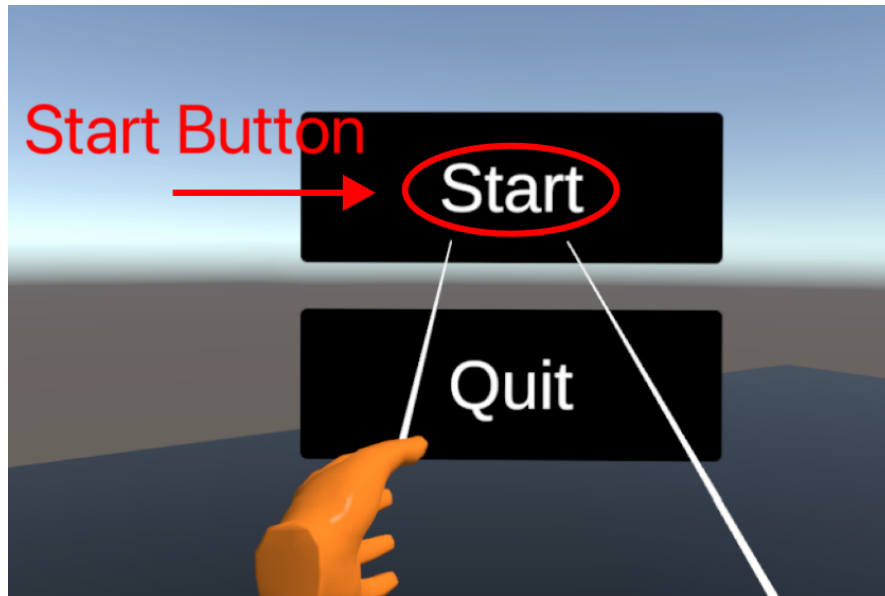
To install and run our game, you need to follow the following steps:

1. Download our compressed file `GRPSoftware.Team16.zip` and extract it.
2. Open the folder `/GRPSoftware.Team16/Builds`.
3. Prepare your VR headsets and connect them to the PCs through Oculus or other OpenXR-compatible platforms correctly. Take the Oculus PC app as an example, you need to enable Quest Link inside the headsets.
4. Double-click the application `STEM Game.exe` inside the `Builds` folder on both PCs.



5. After clicking the `STEM Game.exe` file, players will be initialized as Zeke (the fox) and Yuki (the rabbit) .

6. After **both** players are initialized in the *Lobby*, either one can point at the *Start Button* and presses *Trigger to transport two players to the game scene. (*For *trigger*, please refer to next section in this user manual.)

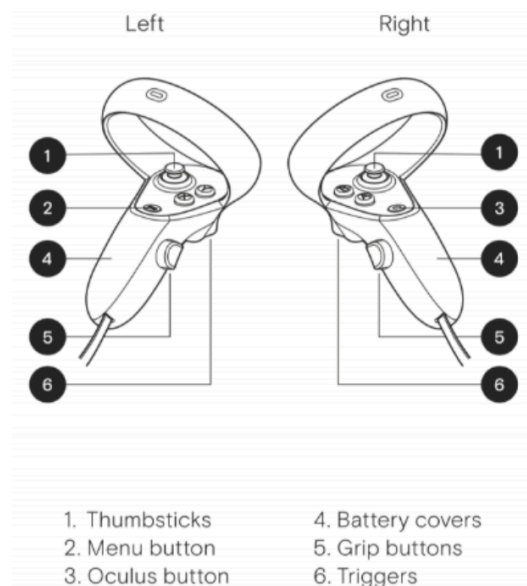


7. Then, let's start and enjoy the game!

Important Notes: To restart the game, make sure that every running instance of the application has been closed, or the server will not allow the new instance to join the same room together.

Controller Interaction Guide

Mastering these controls will enhance your gaming experience, allowing you to move, aim, grab, and perform intricate tasks with ease.



Navigating Your World:

- **Left Thumbstick (L1):** This thumbstick allows you to walk or run in any direction—forward, backward, and sideways. Simply tilt the thumbstick in the direction you wish to move. This intuitive control makes exploring the virtual environment effortless.

Aiming and Looking Around:

- **Right Thumbstick (L2):** This thumbstick controls where you look or aim. By tilting it, you can change your view or aim within the game without the need to physically turn, providing a fluid way to explore and interact with the world around you.

Interacting with Objects:

- **Grip Buttons (L5-R5):** These are your hands in the virtual world. Pressing the grip buttons allows you to grab or interact with objects. Whether you're solving puzzles or engaging with items, these grip buttons are your primary tool for manipulation.
- **Triggers (L6-R6):** For more refined interactions, you will use the triggers. These allow for tasks like acting with 3D UI in the virtual world or performing specific gestures. It's as simple as pressing the triggers to mimic a pinch motion in the game.